Quantitative Action Quality Assessment

Parting the Wild Horse's Mane: form-level sequences

Segmented SMPL Sequences

Reference



Start ... End

Students



Start ... End

For each student

(a) Uniform-Anchor Pose Comparison

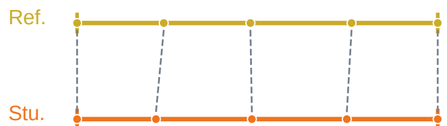
Uniform Sampling

1 FPS

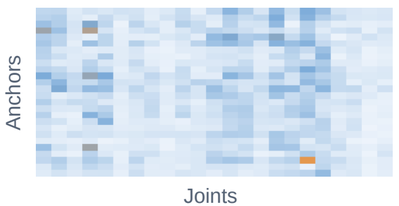


Reference Anchors

SMPL-DTW Matching



Joint Distances

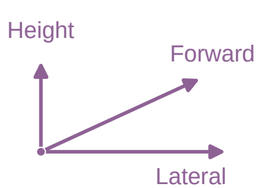


Mean Pose Distance

(b) Pelvis Motion Comparison

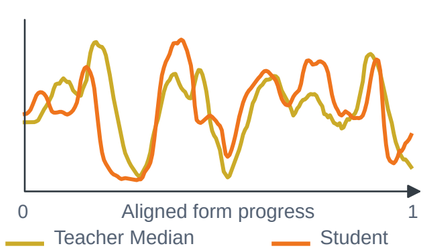
Body-Relative Signals

Pelvis - ankle midpoint



Trajectory Comparison

Lateral pelvis trajectory



Pelvis Deviation

- Horizontal trajectory error
- Endpoint deviation
- Vertical trajectory error

Weighted Sum

Equal-Weight Borda Fusion

Pose Rank

Pelvis Rank

Pose 50% + Pelvis 50%

Relative Ranking

